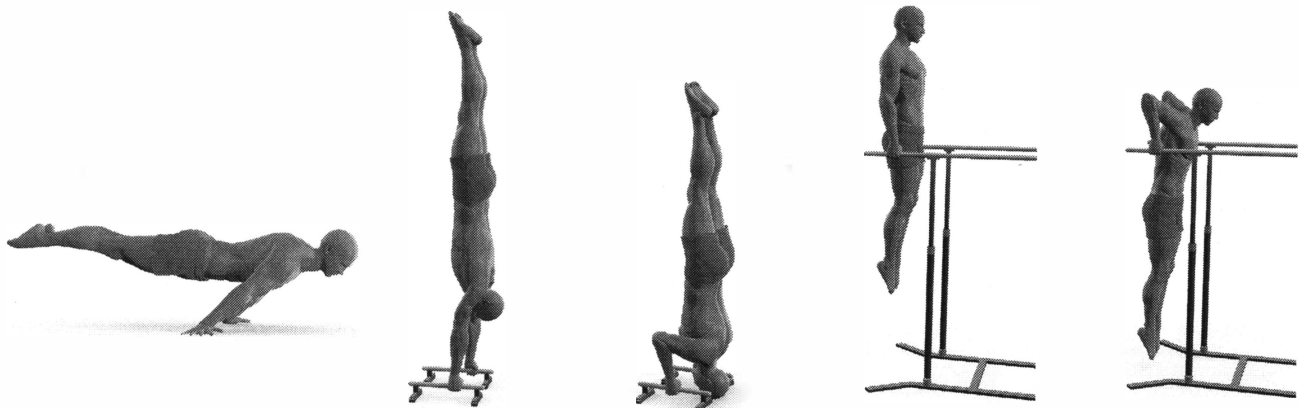


STRUCTURAL BALANCE CONSIDERATIONS

OVERVIEW OF SHOULDER HEALTH

Before you consider selecting exercises to form into a routine that compliments your goals, it is important to consider how certain exercises will affect your body. The most absurd example, of course, is when we look into a typical commercial gym, where you will see many guys performing exercises focused on increasing the aesthetics of their “beach muscles.” The exercises that are most often abused are the bench press and bicep curls. Since this focuses only on the muscles in the front of the body, it may lead to imbalances, which can cause frustration, pain, and injury.

A similar experience from my own training stimulated me into thinking about the issue of structural balance. Back in college, I was training using many pushing exercises, such as planche, dips, and handstand pushups, with only a negligible amount of pulling exercises, such as pull-ups. This caused me to develop pain in the front portion of my shoulder, and also provoked my interest in physical therapy and injury prevention. After researching my injury, I saw that there were errors in my training, which I corrected by over exaggeration of pulling exercises and other rehabilitative work until my shoulders were healthy. As such, this has always been at the forefront of my mind in my own training and in those I coach. I have been informed that there are many other coaches, such as Charles Poliquin, that have talked extensively about this topic, but I am going to share my unique approach to it. If you are interested in some of the other approaches to structural balance, I would encourage you to do your own research.



The shoulder is unique because the small surface area of the joint allows for more range of motion than any other joint in the body. This means it has the capacity for the greatest amount of movement in all directions. However, the small amount of joint contact space also means that any sort of imbalance puts the shoulder at risk for pain and injuries. Thus, it is extremely important to build a routine focused on optimal, balanced shoulder health. After all, how can you train effectively if you are in pain or injured?

In the context of the shoulder, it is important to talk about many of the concepts that I will discuss in regard to injury prevention in later chapters. These concepts are as follows:

- **Range of Motion** – How much movement can be performed at a particular joint.
- **Flexibility** – Best understood as increasing the range of motion by stretching the muscles through relaxation of the nervous system. The most easily understood example of flexibility to increase range of motion is working to obtain the splits.
- **Passive Mobility** – Taking a joint through its range of motion, but without the aim to increase the range of motion, as is the case in flexibility training. Passive mobility is a good way to begin to warm up your joints for a workout. For instance, put your wrists on the floor and move your body over them. This will take your wrists to the edge of the range of motion without contracting the wrist muscles and is a great example of passive mobility.
- **Active Mobility/Flexibility** – These two terms are often used interchangeably, but I typically use the term active flexibility. In reality, it's mostly active mobility, since range of motion must be gained through flexibility training prior to actual use. Once this is accomplished, using the new range would be called mobility training. An example of this would be using something like a pike or straddle stretch to improve compression, where you are actively engaging your abdominals and hip flexors to move your face close to your knees or the ground. Likewise, working a standing split or kick with your leg above your head is an example of active mobility or flexibility.
- **Stability** – A type of training that focuses primarily on neuromuscular re-education, which entails teaching the body to re-learn how to move itself properly. For example, when recovering from an ankle sprain, one of the primary methods of regaining body awareness (proprioception) in the foot and ankle is to work on single leg balances. This activates the muscles around the ankle to stabilize it from wobbling and retraining your balance. Another example would be using planks as a core stability exercise, in order to train the body to keep tension in the core, which will result in the ability to hold itself still against gravity.

My first axiom in bodyweight training is as follows: *Keeping the shoulders and shoulder blades operating optimally is the key to bodyweight strength success.*

Anatomically, the shoulder is composed of the *glenohumeral joint* and the *scapulothoracic joints*. (The layman's terms of which are the *shoulder* and *shoulder blades*, respectively.) Shoulder strength can be thought of as a combination of the stability of the shoulder blade against the body and the movement of the shoulder to exert force. The muscles surrounding the scapula keep it stabilized against the body, which provides a solid foundation for the shoulder to exert force. The muscles around the shoulder then provide the force to execute exercises. If you throw any of these out of whack through training that does not promote proper balance of the shoulder blade muscles and the shoulder itself, you may incur pain, injury, and plateaus in your training.

This is not to say we are going to ignore the elbows, wrists, and rest of the joints in the upper body. Rather, focusing on the shoulder will allow us to correctly select exercises that will build a properly balanced routine. The elbows, wrists, and rest of the arm will naturally flow from this.

My second axiom of bodyweight training builds off of this point: *The shoulder is the lynchpin of the upper body, just as the hip is for the lower body.*

All upper-body movements go through the shoulder. For this reason alone, I believe most of the exercise selection of a routine should be based upon the different movements that occur at the shoulder. Bodyweight movements have a unique quality that sets them apart from typical barbell exercises. This unique quality is that many of them require excellent upper-body flexibility and mobility to perform. For example, proper handstands require the overhead 180 degree shoulder mobility and strength in that position.

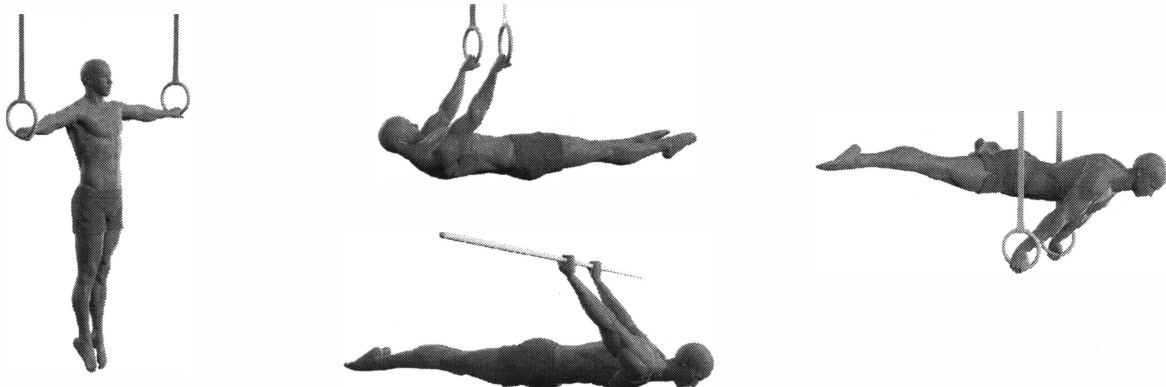
MAINTAINING STRUCTURAL BALANCE

In the first edition of *Overcoming Gravity*, I talked about a simple method of maintaining structural balance and a complex classification of exercises to maintain structural balance. However, this led to mass confusion about what exactly fit in particular categories. Unfortunately, the complex method relied on a good understanding of anatomical joint motion. This turned out to be extremely ineffective for teaching beginners, and made construction of a routine unnecessarily complicated in light of the vast amount of information already in the book. Therefore, I have decided to eliminate the complex method, and revise the simple method in order to make the concepts clearer and simpler to understand.

The simplest method to maintain shoulder structural balance involves utilizing pull and push exercises, which will offset one another. This will allow you to maintain a healthy balance of strength and hypertrophy at the shoulder. This system works for bodyweight exercises in the vast majority of cases. Pulling and pushing exercises are defined below.

- A *pulling exercise* is any exercise in which the center of mass of the body and the hands are moving or pulling *toward* each other.
- A *pushing exercise* is any exercise in which the center of mass of the body and the hands are moving or pushing *away* from each other.

The primary isometric pulling positions that most bodyweight athletes desire to learn include the back lever, front lever, and iron cross. The primary static pushing positions include the planches and inverted cross. The maltese and victorian make up the borderline between pushing and pulling. This is fitting because they are full-body tension exercises to the highest degree, requiring both pushing and pulling muscles to be activated simultaneously.



These positions are illustrated from 12:00 with the front lever, 3:00 with the straddle planche, 6:00 with the back lever, and 9:00 with the iron cross. In the front lever, you can see that the hands will be pulling down toward the center of the range of motion of the body. This indicates a pulling exercise. We can also confirm this by the fact that it primarily works the pulling muscles of the body—such as the latissimus dorsi, teres major, and pectoralis major (lats, teres major, pecs). Likewise, in the back lever and iron cross, the arms are pulling down toward the center of mass the entire time. Conversely, in the planche, the hands are pushing down and away from the body in order to hold it above the rings.

Furthermore, pushing and pulling exercises can be distinguished into two separate categories: vertical/horizontal pushing and vertical/horizontal pulling.

- **Vertical Pushing** – This includes two sets of movements, each at the opposite end of the range of motion—such as handstands, overhead presses, handstand presses and dips.
- **Horizontal Pushing** – This includes several different types of pushup variations and planche variations.
- **Vertical Pulling** – This includes pull-ups and inverted pull-up variations, as well as isometrics—such as iron crosses.
- **Horizontal Pulling** – This includes any and all rowing variations, as well as isometrics—such as back lever and front lever.

Generally, most people will see there are two types of vertical pushing exercises and be confused about which one they should choose: dips or handstand pushup variations. For most new athletes, I suggest beginning with dips, until a greater level of proficiency in strength is attained. Fundamentally, dips are one of the best upper-body pushing exercises for brute strength, and an excellent comparison would be that they are similar to an upper-body squat. Dips will help to build primary pushing strength and overall muscle volume, which ensures a good base for further exercise. Since handstands are usually included as skill work in most beginner routines (in order to familiarize athletes with upside-down and inverted motions), handstands are not neglected from training entirely. Thus, the routine is still balanced, while providing experience for the new athletes in all ranges of motion of the shoulder.

One exception is that there are a fair amount of straight-arm techniques in bodyweight strength training over barbells. I want to clarify the methods of integrating these different planes of motion when choosing exercises for certain goals. First, there is very little “elbows out” in bodyweight strength training, unlike what you typically see with barbell and dumbbell pressing. The reason for this is that flaring the elbows leads to a lack of control in many of the movements. For example, if you flare the elbows out in handstands or handstand pushups, you will fall over almost immediately during the exercises. The elbows need to be tucked in because the body is the lever that needs to be balanced.



Most people who perform handstand pushups (HSPUs) against a wall—or add them as supplementary barbell work—do them with elbows flared out. While this is a bit stronger (because of the added trapezius involvement at a good angle of contraction) it does not allow proper development of strength and control required to learn a freestanding handstand pushup. “Elbows in” allows the body to struggle to balance the freestanding handstand pushup, as opposed to just falling over.

Additionally, there are some exceptions within certain progressions, such as wide-arm pull-ups, wide-arm rows, wide-arm dips, and the iron cross—where the elbows or arms will go out wide. These are important to note in exercise only, as they represent stepping stones to higher progressions in most cases, with the exception of the iron cross. (The iron cross can be thought of as a whole different progression and needs to have specific training for it, which I have highlighted on the chart. Arms wide requires rigorous training of the connective tissue, as the potential for injury is higher without said training.)

THE WHY OF STRUCTURAL BALANCE

As you may know, in typical routine construction for beginners, there is often a focus on the “beach muscles.” For example, five different variations of bench press and curls may dominate a routine. This is poorly constructed routine because it is composed primarily of pushing exercises. The bench press will add strength and hypertrophy to the chest and anterior shoulder, but this routine fails to strengthen the back of the shoulder, which is critical to maintaining structural balance.

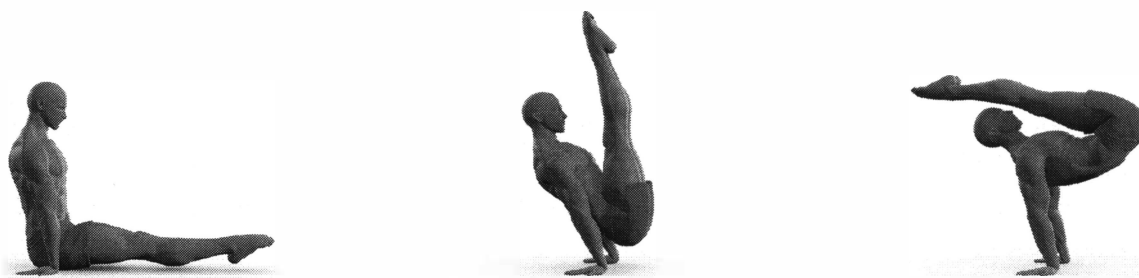
If an athlete persists in this type of poorly-constructed routine for long, they will end up with poor posture, tightness in the front of their chest, and injuries to their rotator cuff or other structures within the anterior shoulder. In bodyweight and barbell training, most routines lack a proper amount of pulling exercises. Pulling movements, particularly horizontal pulling, requires the development of three areas of the back, which are often overlooked. These areas are the scapular retractors, posterior deltoids, and external rotators. These muscles help stabilize the back of the shoulder blades and shoulder, respectively, and are heavily worked in horizontal pulling (rowing) motions.

Poorly constructed barbell routines will lead to improperly developed posterior strength of the shoulder, which can result in pain or injury. If you are experiencing an onset of clicking, popping, or cracking in your joints (where there was none previously), this may indicate that your structural balance is not correctly maintained, especially if your routine has inconsistent amounts of pulling or pressing, resulting in uneven shoulder muscle distribution. For example, if the anterior shoulder becomes too strong, the head of the humerus may start impinging (strongly rubbing or pinching against) and clicking across the anterior portion of the shoulder socket. In this case, the anterior portion of the shoulder labrum may be the part of the shoulder that is clicking or popping.

The shoulder labrum is a piece of cartilage that functions similar to the meniscus (in the knee), so it should be obvious why you do not want to construct a routine that has a high potential for injury in that area. Further imbalances may start to wear down the labrum, leading to potential tears or intractable damage. However, clicking or popping sounds may not always indicate such a problem. If you have always had clicking, it is likely nothing to worry about, though it may be an issue if it becomes louder or there is an onset of pain that develops simultaneously to performing an imbalanced routine.

It is not good for the muscles or joints of the body to get out of balance. Your body is much more intelligent than you may realize, as your nervous system is constantly aware of sensation and proprioception feedback, which causes the body to limit force development and compensate as necessary. In other words, if your body senses there is an imbalance that could result in injury, it will limit strength and musculature development in the surrounding areas, and may begin compensation patterns to prevent injury. This is easily avoided by keeping your exercise routine properly balanced.

I talk about potential injury issues, not to scare you away from exercising or performing bodyweight exercises, but to help you work toward your goals both *safely* and effectively. Exercising and becoming stronger is critical to developing a healthy body, but there is no point in getting strong if you are injured. You must take this into account when constructing a routine, especially one that is based in bodyweight exercises that are tough on your joints and connective tissues.



Pictured is the manna. In my opinion, the reason why the L-sit/V-sit manna progression is so important is because it effectively counteracts the effect of a heavy pushing routine with one simple series of exercises. This set of progressions works the scapular retractors, posterior deltoids, and external rotators without having to do two or three different exercises, thus saving much time and effort. Plus, you get to work toward an awesome skill.

GENERAL RECOMMENDATIONS

There are a diverse amount of exercises in bodyweight training—the only limits are your individual creativity. That being said, there is a small selection of exercises that I highly recommend for your training program. You should know that these exercises are recommended for important reasons.

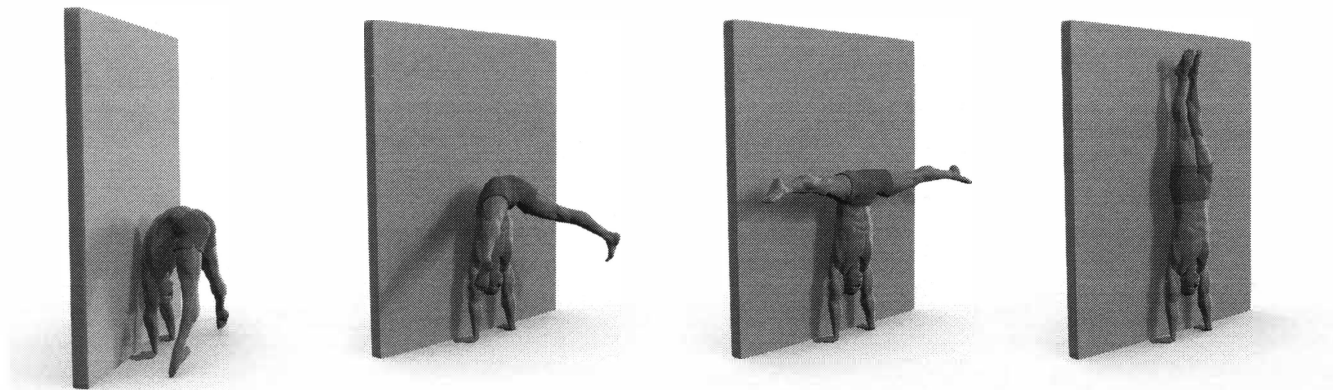


Coupling Handstands and the L-Sit/V-Sit Manna Progression

In a perfect world, everyone would work both handstands and the manna as coupled skills. I like this for multiple reasons:

- Development of strength in active flexibility positions is the key to dominating bodyweight movements. These will drastically increase your body awareness and ability to control your muscles through all ranges of motion.
- Handstands work the overhead range of motion of the shoulders, and manna works the limit of the backward range of motion of the shoulders. Working stability and balance at the ends of the range reinforces strength within the rest of the range and protects the joint from injury.
- Training for the manna progressions requires more-than-adequate flexibility to be developed in the shoulder girdle and legs.
- Both handstands and manna have built-in core control and strength work. Thus, less time needs to be spent on core conditioning, and more emphasis can be placed on skill and strength development.
- As previously mentioned, developing these skills simultaneously will ensure that imbalances of the shoulder will be less likely to develop.

The alternative is additional scapular retraction work or horizontal pulling exercises that focus on pulling the elbows back, in order to squeeze the shoulder blades together, which keeps the pulling and pushing exercises balanced. However, this is extra work compared to the benefits of utilizing this coupled progression.



Straight-Arm Press Handstands

These are at the core of gymnastics. They require incredible upper body and core control, which will greatly benefit your training as you learn them and in the intermediate phase. They will also help as you move into the more advanced stages.

Straight-arm press handstands from support are actually a very good supplemental exercise for planche training because they require very solid amounts of shoulder girdle strength moving through a large range of motion. They extensively work the scapular stabilizers—especially the upper trapezius, lower trapezius, and serratus anterior—when going overhead. While this means nothing to the layperson, the triangle of forces that these three muscles exert on the scapula are key in both overhead pressing and the maintenance of proper scapular stability against the thoracic cage in posture and movement.

Straight-arm press handstands are an extension that will help you master the handstand, as you can easily hold handstands once you can control them from a straight-arm press. Plus, they look cool. This principle also applies to specific bent-arm pressing. Once you master straight-arm press handstands (straddle press at the very least), most athletes will be able to do a bent-arm press as well. The same is not true in the opposite direction, which is why this skill must be developed first or concurrently.

Handstand pushups, rope climbing (if available), and full back lever development are very important to the process of developing high-level strength, though not required. The reasoning behind this is they not only help with the development of strength, but also with joint integrity and building up connective tissue. Thus, they are likely to help significantly in the development of high-level skills. Indeed, I have some of these exercises as prerequisites to other high-level strength progressions on the charts and in the exercise section. I only have a small section on rope climbing in the exercise section, but if you have access to a rope you should definitely utilize it.

Just remember to organize your workouts so that you can keep your body structurally balanced and injury free. The rest is really up to you and what you want to accomplish. Most importantly, have fun!

STOP READING. TAKE ACTION.

1. Organize your goals into categories of push and pull. I have already significantly helped you in this process by placing exercises on the particular charts.
2. Break down the push and pull exercises into categories of horizontal and vertical for each.
3. If there is a disparity in goals between any of the categories, I would suggest that you either supplement with barbell work in that area (which I will not cover in this book), train the progressions that I recommend, or selecting exercises to shore up those weak points.

CHAPTER 4 SUMMARY

STRUCTURAL BALANCE CONSIDERATIONS

Knowledge Base

It is important to keep structural balance considerations in mind, in order to keep your shoulders healthy. The shoulders are the lynchpin for the upper body in terms of strength development. Therefore, it is of utmost important to construct your entire routine around maintaining healthy shoulders.

Application

I prefer the push and pull system for classifying different types of bodyweight exercises, as it is simple and effective. The basics of this system can be summarized as:

- Pick horizontal and vertical pushing exercises. For newbies, the handstand covers vertical pushing upward, so choose a vertical pushing downward.
- Pick a vertical pulling exercise and one horizontal pulling exercise.
- Put the L-sit progression to manna and handstand in your routine.

The handstand will be included as skill work and overhead facilitation. The L-sit will be included as core work and opposition to the handstand, in order to progress toward manna. The other pushing and pulling movements in the main exercises are going to be balanced. Add in at least two leg exercises, either barbell or bodyweight.

It is best if your vertical upward pushing starts out as handstand work, as handstands are critical for the development of body proprioception and control. Progression in this skill signifies the level of ability of the user. Very few people develop strong bodyweight abilities without proper handstand work.